

# Acemoglu & Restrepo (2018)

The race between automation and new tasks

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[github.com/amchavezu/ai-06-restrepo](https://github.com/amchavezu/ai-06-restrepo) NBER Working Paper 22252, revised June 2017. Proposition numbering follows this

version.

# The aggregate economy becomes a task allocation problem

**PAPER** Final output combines a unit measure of tasks:

$$Y = \tilde{B} \left( \int_{N-1}^N y(i)^{\frac{\sigma-1}{\sigma}} di \right)^{\frac{\sigma}{\sigma-1}}.$$

- ▶  $I$  is the automation frontier.
- ▶  $N$  is the frontier of newly created tasks.
- ▶  $n = N - I$  measures the task range left to labor.

## Automation

Capital can perform more existing tasks.

## Reinstatement

New tasks expand the set in which labor has comparative advantage.

**INTERPRETATION** The unit of analysis is the aggregate task frontier, not an individual worker.

Source: Acemoglu and Restrepo, NBER revision, Sections 1 and 2.1

# Effective costs determine who performs each task

**PAPER** All tasks can use labor. Tasks  $i \leq I$  can also use capital.

$$p(i) = \begin{cases} \min\{R, W/\gamma(i)\}^{1-\eta}, & i \leq I, \\ (W/\gamma(i))^{1-\eta}, & i > I. \end{cases}$$

**Cost threshold**

**Equilibrium threshold**

$$\frac{W}{R} = \gamma(\tilde{I})$$

$$I^* = \min\{I, \tilde{I}\}$$

**INTERPRETATION** Assumption 1 makes  $\gamma(i)$  strictly increasing. Capital takes the low-index tasks and labor takes the high-index tasks.

Source: Equations (5) and (6), NBER pp. 7–8

# The task threshold becomes an endogenous production share

**PAPER** Under Assumptions 1, 2, and 3, Proposition 1 gives a unique static equilibrium:

$$Y = \frac{B}{1-\eta} \left[ (I^* - N + 1)^{1/\hat{\sigma}} K^{\frac{\hat{\sigma}-1}{\hat{\sigma}}} + \left( \int_{I^*}^N \gamma(i)^{\hat{\sigma}-1} di \right)^{1/\hat{\sigma}} L^{\frac{\hat{\sigma}-1}{\hat{\sigma}}} \right]^{\frac{\hat{\sigma}}{\hat{\sigma}-1}}.$$

- ▶ Raising  $I^*$  increases capital's task share.
- ▶ Raising  $N$  creates a task at the top and removes one from the bottom.
- ▶ The factor shares change because technology reallocates tasks.

Source: Proposition 1 and equation (12), NBER pp. 9–10

# Automation and new tasks push labor in opposite directions

**PAPER** In the technology-constrained region  $I^* = I < \tilde{I}$ :

|                     | More automation $dI > 0$ | More new tasks $dN > 0$ |
|---------------------|--------------------------|-------------------------|
| Relative wage $W/R$ | Falls                    | Rises                   |
| Labor share         | Falls                    | Rises                   |
| Employment          | Falls                    | Rises                   |
| Productivity        | Rises                    | Rises                   |
| Wage level $W$      | Ambiguous                | Rises                   |

**INTERPRETATION** Productivity growth does not identify who receives the gains. The task margin separates the wage from output and the labor share.

Source: Propositions 2 and 3, NBER pp. 11–13

# The wage sign is a race between two exact terms

Set  $dN = 0$  in Proposition 3:

$$\frac{d \ln W}{dl} = \underbrace{\frac{B^{\hat{\sigma}-1}}{1-\hat{\sigma}} \left[ \left( \frac{W}{\gamma(I^*)} \right)^{1-\hat{\sigma}} - R^{1-\hat{\sigma}} \right]}_{P_I \text{ productivity}} - \underbrace{(1-s_L) \frac{\Lambda_I}{\hat{\sigma} + \varepsilon_L}}_{D_I \text{ displacement}}.$$

Automation raises the wage if and only if  $P_I > D_I$ .

**DERIVATION**  $P_I$  measures the cost saving at the marginal automated task.  $D_I$  measures the pressure from concentrating labor in fewer tasks.

**INTERPRETATION** When the effective wage  $W/\gamma(I^*)$  is close to  $R$ , the productivity gain is small and displacement dominates.

Source: Proposition 3 and discussion, NBER pp. 12–14

## A numerical check makes the threshold visible

**DERIVATION** Normalize  $R = B = 1$  and use illustrative values  $\hat{\sigma} = 1.5$ ,  $s_L = 0.6$ ,  $\varepsilon_L = 0.5$ , and  $\Lambda_I = 2$ .

| Effective cost ratio $c$ | $P_I$ | $D_I$ | $d \ln W / dI$ |
|--------------------------|-------|-------|----------------|
| 1.10                     | 0.093 | 0.400 | -0.307         |
| 1.5625                   | 0.400 | 0.400 | 0.000          |
| 2.00                     | 0.586 | 0.400 | 0.186          |
| 4.00                     | 1.000 | 0.400 | 0.600          |

**INTERPRETATION** The sign changes at  $c = (W/\gamma(I^*))/R = 1.5625$ . This table evaluates the decomposition. It is not a calibration of the economy.

Source: Own computation in `analysis/wage_decomposition.py`

# Capital adjustment changes the long-run wage result

**PAPER** Along an interior BGP, the Euler equation pins down

$$R = \rho + \delta + \theta g.$$

For  $n > \bar{n}(\rho)$ , a permanent increase in automation lowers  $n = N - l$ .

| Outcome     | Short run        | Long run                              |
|-------------|------------------|---------------------------------------|
| Wage        | May fall or rise | Rises                                 |
| Employment  | Falls            | Falls                                 |
| Labor share | Falls            | Falls                                 |
| Rental rate | Rises            | Returns to $\rho + \delta + \theta g$ |

**INTERPRETATION** The wage can recover because capital accumulates. That recovery does not restore employment or labor's income share.

Source: Proposition 5 and Figure 6 discussion, NBER pp. 18–20

# Scientists choose the direction of technology

Scientists allocate effort to automation or new-task creation:

$$\dot{I} = \kappa_I S_I, \quad \dot{N} = \kappa_N S_N, \quad S_I + S_N \leq S.$$

The task gap moves according to relative innovation values. An interior BGP requires

$$\kappa_I v_I(n) = \kappa_N v_N(n).$$

## Automation value

Cost saving from replacing labor with capital at the bottom of labor's task range.

## New-task value

Cost saving from replacing the capital-intensive alternative with labor at the frontier.

Source: Equations (22), (28), and (29), NBER pp. 21–25

## Proposition 6 maps four balanced-growth regimes

**PAPER** Assume  $\gamma(i) = e^{Ai}$ , Assumption 2,  $\hat{\sigma} > \zeta$ , and  $S < \bar{S}$ .

| Parameter region                                                           | BGP task gap               | Outcome             |
|----------------------------------------------------------------------------|----------------------------|---------------------|
| $\rho < \bar{\rho}$                                                        | $n = 0$                    | Full automation     |
| $\rho > \bar{\rho}, \kappa_I/\kappa_N > \bar{\kappa}$                      | $n \in (\bar{n}(\rho), 1)$ | Unique interior BGP |
| $\rho > \bar{\rho}, \underline{\kappa} < \kappa_I/\kappa_N < \bar{\kappa}$ | Interior                   | Multiple BGPs       |
| $\rho > \bar{\rho}, \kappa_I/\kappa_N < \underline{\kappa}$                | $n = 1$                    | No automation       |

**INTERPRETATION** The “race” has an interior solution only for part of the parameter space. Labor redundancy remains a genuine equilibrium possibility.

Source: Proposition 6, NBER pp. 25–27

# The interior race is stable because prices redirect research

At the unique interior BGP,

$$\kappa_I v_I(n) = \kappa_N v_N(n).$$

- ▶ A fall in  $n$  means automation has run ahead.
- ▶ The resulting factor-price response lowers the return to further automation.
- ▶ New-task research becomes relatively more attractive.
- ▶ The task gap returns to its interior BGP when the unique-equilibrium condition holds.

**PAPER** With  $\theta = 0$ , the interior BGP is globally saddle-path stable. With  $\theta > 0$ , it is locally unique and asymptotically saddle-path stable.

**INTERPRETATION** A temporary automation shock can unwind. A permanent increase in  $\kappa_I/\kappa_N$  instead moves the economy to a new BGP with lower employment and labor share.

# My analytical and computational work

## Analytical check

- ▶ Isolated the exact wage condition  $P_I > D_I$ .
- ▶ Checked the signs rather than repeating the headline that automation hurts labor.
- ▶ Separated the short-run fixed-capital result from the BGP response.

## Computational check

- ▶ Evaluated Proposition 3 over a grid of marginal cost ratios.
- ▶ Recovered the break-even ratio  $c = 1.5625$  for the illustrative parameters.
- ▶ Kept the exercise distinct from a calibration of the paper.

## Extension sketch

Add separate policy wedges to  $\kappa_I v_I(n)$  and  $\kappa_N v_N(n)$ , then study how innovation policy shifts the interior BGP.

# The Lean workflow preserves the boundary between source and proof

1. Pin the NBER June 2017 source bytes and inventory the named claims.
2. Translate each selected result into one transparent source-facing Spec.
3. Keep the Lean proof endpoint separate from the semantic statement.
4. Run the paper-scoped build and audit checks.
5. Report partial boundaries instead of turning missing proofs into assumptions.

**Paper folder:** AR18RaceManMachine

**Required check:** `python3 scripts/paper_contribution.py check`  
`AR18RaceManMachine -fast`

Source: `AppliedModelingLib` paper-formalization workflow

# Required Lean slide: source claim, statement, and proof

## Original mathematical claim

Selected equation or proposition will be inserted from the generated paper folder.

## Lean statement and proof fragment

```
-- LEAN_CODE_PLACEHOLDER  
-- Replace only with exact generated code from AR18RaceManMachine.
```

## What Lean verifies

The final version will name the exact Spec, identify the assumptions and conclusion, explain one nontrivial proof line, and state whether the source claim is exact or conditional.

# Where I did not believe the AI

## Claim to test

“Automation necessarily reduces wages and the labor share.”

## Verdict

The labor-share statement is correct in the technology-constrained static region. The wage statement is incomplete.

$$\frac{d \ln W}{dI} = P_I - D_I$$

Automation raises the wage when cost savings are large enough. On the interior BGP, capital adjustment makes the long-run wage rise even as employment and the labor share fall.

Owner-supplied photograph still required: [hand/wage-derivation.jpg](#)

Source: Own hand derivation of Proposition 3; NBER pp. 12–14

# What the model changes about the automation debate

- ▶ Productivity gains do not determine the wage, employment, or labor share by themselves.
- ▶ Automation displaces labor from existing tasks. New tasks reinstate labor in the task frontier.
- ▶ The short-run wage effect is conditional on productivity gains relative to displacement.
- ▶ An interior innovation race can be stable, but full automation remains possible.
- ▶ Lean is useful here because every domain restriction and inequality must be stated explicitly.

[github.com/amchavez/ai-06-restrepo](https://github.com/amchavez/ai-06-restrepo)